

IMMUNE THROMBOCYTOPENIA (ITP)

MANAGEMENT IN THE CHILDREN'S EMERGENCY

1. BACKGROUND

ITP is an acquired autoimmune disorder characterized by a low platelet count from increased platelet destruction & impaired production. Most cases are idiopathic. There are viral and environmental triggers, or it may be 2' to an underlying immunologic defect.

This clinical guide focusses on the initial evaluation & management of **newly diagnosed primary ITP**.

1.1 EPIDEMIOLOGY

- Peak incidence at 2 to 5 years of age; Boys > Girls (1.2 : 1)
- The rate of spontaneous remission at 1-year is 62 – 74 % (Bennett CM, 2018) (Imbach P, 2006)

1.2 CLASSIFICATION OF ITP

Primary ITP		Secondary ITP
Newly diagnosed	Within 3 months of symptom onset	Immune-mediated with underlying cause (e.g., drug-induced, SLE, HIV infection)
Persistent	Between 3 to 12 months after diagnosis	
Chronic	> 12 months after diagnosis	

2. DIAGNOSIS

ITP is a diagnosis of exclusion.

	Details
Typical Presentation	Acute onset petechiae/bruising in an otherwise well-appearing child. Preceded by illness or vaccination (within the past month). Lack of systemic signs & symptoms to suggest another cause of thrombocytopenia. For example: - Enlargement of lymph nodes, liver, or spleen - Features of Fanconi anemia i.e., abnormal thumbs or forearms ± hyper- or hypopigmented skin lesions - Longstanding history of thrombocytopenia - History of atypical bleeding / bleeding diathesis - Family history of thrombocytopenia or bleeding of unknown etiology
FBC features	Platelet count < 100 x 10⁹ /L Rest of FBC normal with a normal peripheral smear (no blasts, no hemolysis).

3. SEVERITY OF BLEEDING

3.1 MODIFIED BUCHANAN AND ADIX BLEEDING SCORE

Severity of ITP is based off clinical bleeding criteria and not the patient's platelet count.

Grade	Severity	Description
0	None	No bleeding
1	Minor	Petechiae (≤ 100), ≤ 5 bruises measuring ≤ 3 cm in diameter, no mucosal bleed
2	Mild	Petechiae (> 100), > 5 bruises measuring > 3 cm in diameter, no mucosal bleed
3	Moderate (low risk)	Epistaxis (< 5 mins), oral/palatal petechiae, painless oral purpura
	Moderate (high risk)	Epistaxis (≥ 5 minutes), hematuria, painful oral purpura, menorrhagia, hematochezia
4	Severe	Organ involvement (e.g., muscle, joint)
5	Life-threatening/fatal	ICH, life-threatening or fatal hemorrhage at any site

4. MANAGEMENT

- Admit if platelet $< 20 \times 10^9/L$ and any of the following:
 - Exposed to anti-platelet (e.g., NSAIDs) and/or anti-coagulant medication (e.g., heparin, warfarin)
 - Concomitant bleeding disorder.

Grade	Severity	Management
0	None	○ Discharge[#] with red flags & precautions. ○ No corticosteroids recommended. ○ Early review with Paeds Haem within 24 to 48H for review of blood film
1	Minor	
2	Mild	
3	Moderate (LR)	○ Discharge[#] with red flags & precautions. ○ Early review with Paeds Haem within 24H for review of blood film ○ Corticosteroids recommended after Haem review. - Prednisolone <u>2 – 4 mg/kg</u> (max. 160mg) daily for max. 7 days, or - Dexamethasone <u>0.6 mg/kg</u> (max. 40mg) daily for 4 days.
3	Moderate (HR)	○ Admit for observation, CRIB
4	Severe	
5	Life-threatening	

[#] Discharge only if admission criteria not met.

4.1 DISCHARGE ADVICE

- Avoid NSAIDs
- Avoid contact sports
- Avoid IM injections
- Return for evaluation following head injury/trauma
- Red flags for raised intracranial pressure: headache, nausea, vomiting, drowsiness, weakness, seizures

4.2 ARRANGING FOLLOW UPS

4.2.1 ORDERING TCUS

“Dispo” activity tab → “New Orders” → Search for “TCU Paediatric medicine – NUHS”

TCU PAEDIATRIC MEDICINE - NUHS

☒ Accept
 ☒ Cancel

Services

Please select :

Location:

Reason for request

Does patient require lab?

Status:

Expected Date:

☐ Approx.

Comment:

Expires:

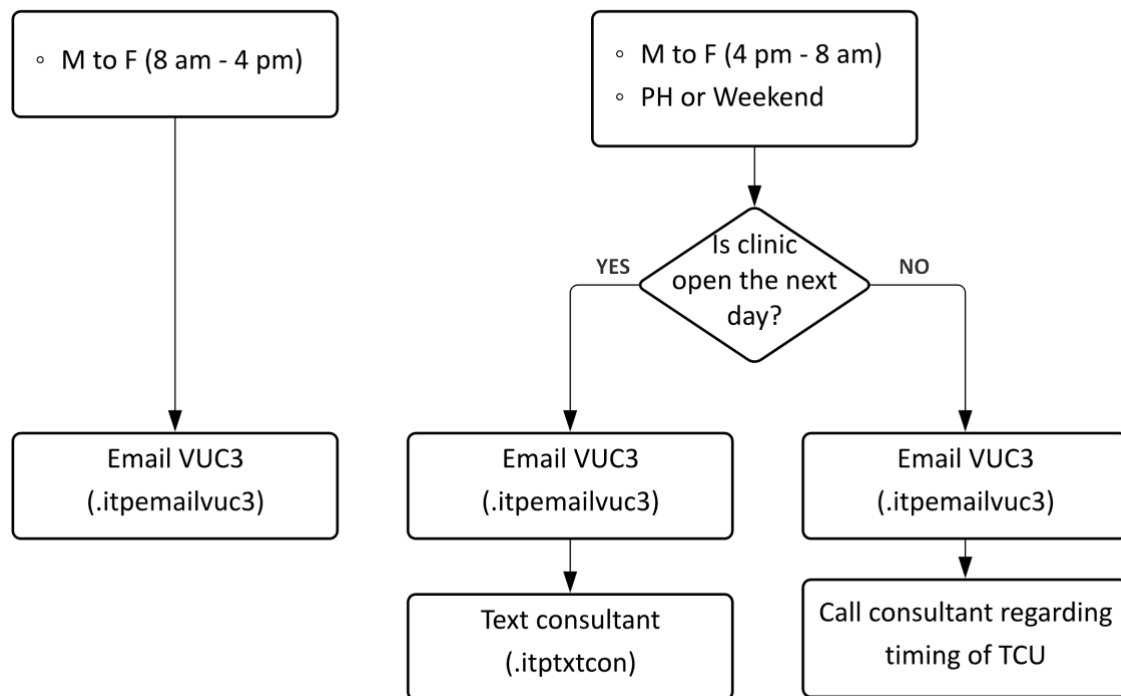
Comments:

Force book as per ITP guidelines

☒ Accept
 ☒ Cancel

Services:	Oncology
Reason for request:	ITP
Expected Date:	Tomorrow Uncheck ☐ Approx.
Comments:	“Force book as per ITP guidelines”

4.2.2 ARRANGING URGENT FOLLOW UPS



Monday to Friday; 8am to 4pm	Monday to Friday; 4pm to 8am, Public holiday or weekend, Department ad hoc half/full day
- Email VUC3 – vuc3@nuhs.edu.sg - Smartphrase template: .itpemailvuc3 - Subject: “Urgent TCU from CE (ITP)”	<u>If clinic is open the next day:</u> - Email VUC3 + cc consultant on service - Text consultant (smartphrase template: .itptxtcon) <u>If clinic is closed the next day:</u> - Email VUC3 as above + cc consultant on service - Call consultant regarding timing of TCU

5. REFERENCES

- Bennett CM, N. C. (2018). Predictors of remission in children with newly diagnosed immune thrombocytopenia: data from the Intercontinental Cooperative ITP Study Group Registry II participants. *65*(1).
- Imbach P, K. T. (2006). Childhood ITP: 12 months follow-up data from the prospective registry I of the Intercontinental Childhood ITP Study " Group (ICIS). *46*(3).
- The American Society of Hematology. (2019). American Society of Hematology 2019 guidelines for immune thrombocytopenia.